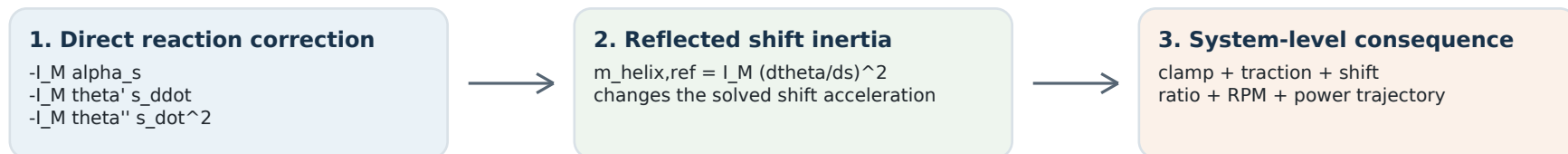


Secondary Helix Dynamics Discovery Checkpoint

CINDER v1.1.2 | Helix-topology studies E1 through E5.8 | 17 September 2026

Current top-level conclusion	What this document preserves
Dynamic secondary-helix coupling changes transient CVT behavior through two separate but coupled mechanisms: (1) explicit acceleration-dependent helix reaction terms, and (2) reflected movable-member inertia in the shift coordinate. Near severe reverse-power/traction transitions, these effects materially change slip timing, backshift, ratio, clamp and power histories.	The study progression, numerical anchors, strongest figures, paired full-vs-quasi-static comparison, interpretation guardrails, and exact restart points. The intent is to let the project resume later without re-deriving what was learned.



The two paths are coupled: reflected inertia changes s_{ddot} , which feeds the direct helix reaction term.

Framing: helix lift-off / opposite-flank demand is an important consequence and a useful diagnostic boundary, but it is not the primary claim. The broader result is that a dynamically modeled helix changes the transient response of the CVT.

Status at checkpoint

- Helix discovery phase is considered complete. No additional broad sweep is recommended before synthesis/write-up.
- A small nearby-case replication set may be useful later only if final figures need robustness evidence.
- No literature-priority claim is made here. These are model/results conclusions, not a claim that no prior model can partially predict the same mechanisms.

1. Executive findings

A. Dynamic helix terms are physically active even before slip

E5.6 reached a pre-traction total dynamic correction of -1.121 N m and a shift-acceleration contribution of -0.926 N m. E5.7 stock 300 deg control still showed a -0.891 N m shift term while comfortably sticking.

B. The dynamic reaction can change the required helix flank

E5.7 found 14 cases where the trajectory-frozen quasi-static margin remained positive while the full dynamic margin crossed negative. In the selected 270 deg reversal case, the crossing had $M_{h,QS} = +1.811$ N m and dynamic correction = -1.811 N m.

C. Reflected helix inertia is a separate dynamic effect

Near the paired reversal event, the full formulation carried about 7.48 kg of helix-reflected generalized shift inertia, bringing the direct active generalized shift mass to about 9.67 kg versus about 2.19 kg in the quasi-static-helix variant. This strongly moderates the backshift response.

D. Those local mechanics change system trajectories

In E5.8 the severe reversal pair differed by as much as 2.026 mm in shift position and 0.231 in speed ratio. The quasi-static helix left stick about 4.66 ms earlier and remained non-stick about 4.66 ms longer over the simulated window.

E. Away from the topology boundary, the trajectory effect is small

The stock 300 deg paired control showed only 0.051 mm maximum shift difference and 0.0137 maximum ratio difference. This supports a scale/regime-dependent interpretation rather than a claim that quasi-static helix models are always poor.

F. Helix reversal did not imply belt normal-contact loss

Across 30 helix-crossing E5.7 cases, the minimum interpolated secondary normal resultant was about 827 N and no $N_s = 0$ event occurred at the helix crossing. Helix flank admissibility, traction saturation, and belt normal-contact loss are distinct events.

Recommended one-sentence result

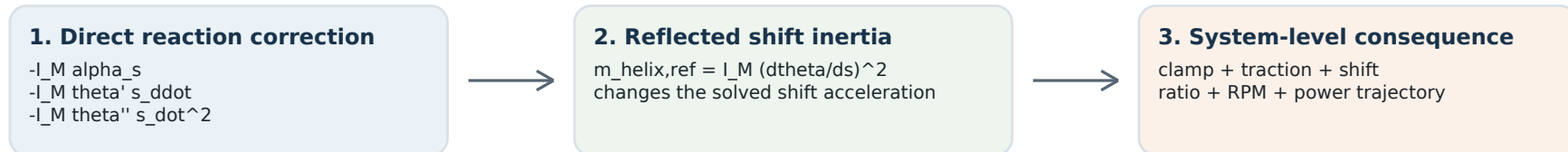
Modeling the secondary helix as a dynamic movable-member mechanism, rather than only a quasi-static torque-to-axial-force map, can alter both the instantaneous helix reaction and the effective shift inertia, producing materially different transient CVT trajectories near severe reverse-power and traction transitions.

2. Physical mechanism and comparator

The production dynamic helix reacted-torque margin used throughout the discovery work is:

$$M_h = f \tau_s + k_{\theta} (\theta_{pre} - \theta) - I_M \alpha_s - I_M \theta'(s) \ddot{s} - I_M \theta''(s) \dot{s}^2$$

with axial helix reaction $F_h = M_h (d\theta/dx_s)$. The trajectory-frozen quasi-static diagnostic/comparator retains the belt-torque and torsional-spring terms but removes the acceleration-dependent movable-member terms.



The two paths are coupled: reflected inertia changes s_{ddot} , which feeds the direct helix reaction term.

Two distinct roles of the same movable-member inertia

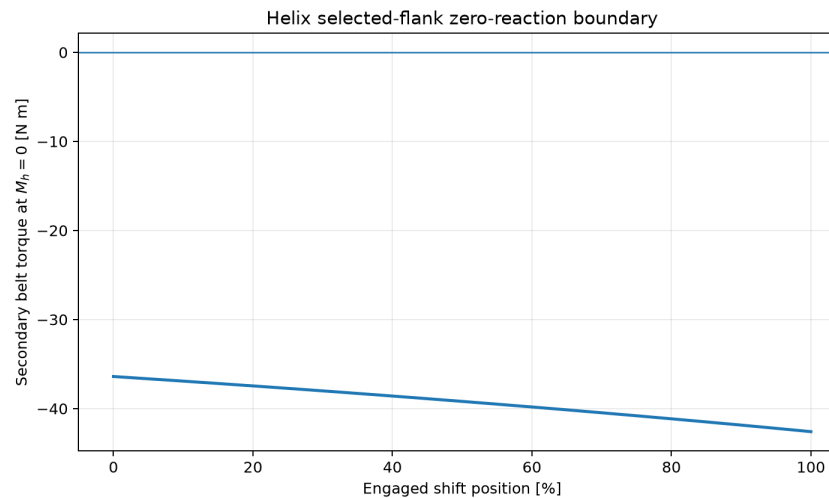
- Direct reaction role: shaft acceleration, shift acceleration and helix curvature contribute directly to the signed helix reaction. The shift term can become large during a hybrid traction transition.
- Reflected inertia role: the movable member contributes approximately $m_{helix,ref} = I_M (d\theta/ds)^2$ to the generalized shift coordinate. This resists rapid shift motion and therefore changes the solved s_{ddot} that feeds the direct reaction term.
- Coupled consequence: the dynamic model is not just a quasi-static model plus an extra force correction. It follows a different trajectory because the generalized inertia itself is different.

Topology and interpretation guardrail

The maintained results model is a zero-clearance bilateral/slotted helix. A negative signed margin means the originally selected flank would be inadmissible and the opposite slot flank is required. E5.8 compares full-dynamic and quasi-static versions of this same signed/slotted topology; it is not a true unilateral detached-contact comparison.

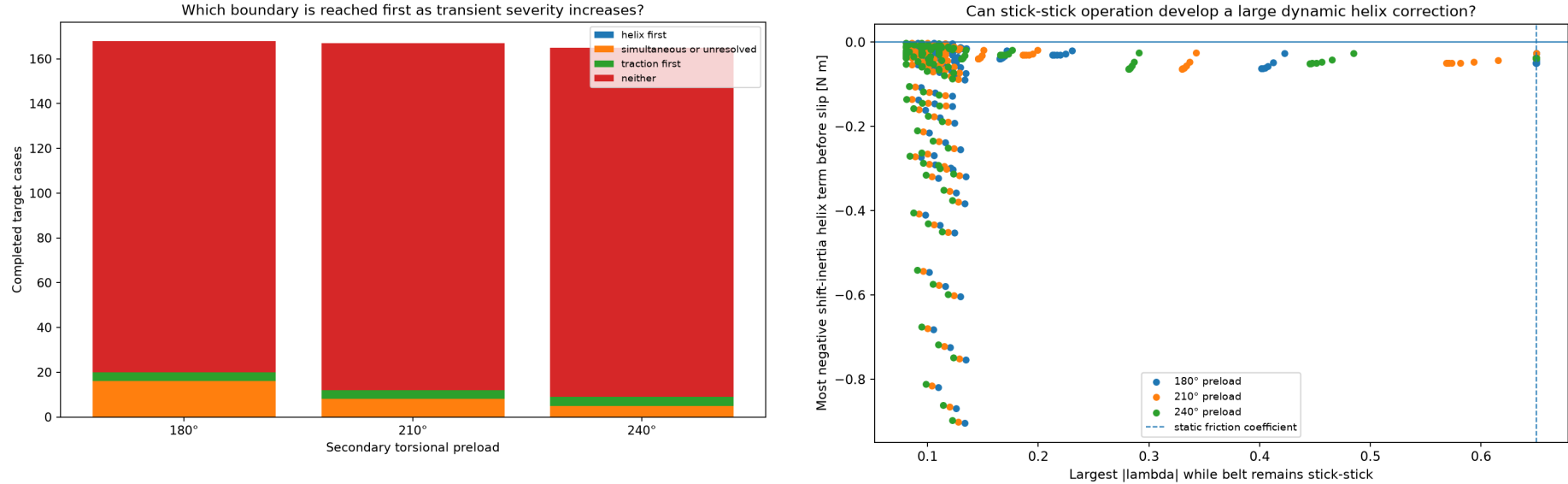
3. Study progression: what each stage established

Stage	Purpose	Main result	Current interpretation
E1 - reaction map	Locate quasi-static zero reaction boundary	Zero-margin secondary belt torque spans -36.40 to -42.59 N m across shift.	Pure geometry/spring boundary; not a vehicle result.
E2 - forward control	Check ordinary forward launch	Minimum margin +19.58 N m; 0 zero crossings.	Normal forward launch is far from flank reversal.
E3 - direct stress screen	Apply broad direct torque stress	99 cases: 90 one-flank, 9 near-boundary, 0 opposite-flank.	Huge torque alone does not guarantee reversal; state and power-flow path matter.
E4 - scenario discovery	Search physical/bench scenario families	67/68 completed. One downhill engine-braking and one bench-backdrive family case required opposite flank.	Reverse-power / backdrive emerged as the relevant neighborhood.
E5 - lift-off envelope	Map thresholds and dynamic-only candidates	185 vehicle cases; 0 reference-scale vehicle lift-off cases at ≤ 28 N m engine braking. 27 strict dynamic-only candidates, 0 clean strict candidates.	Dynamic-only algebra exists, but clean forward-power stick-stick reversal was not found.
E5.5 - forward-power isolation	Deliberately search for clean dynamic-only lift-off	90/90 completed; 0 dynamic-only first crossings, 0 gold cases.	Do not force a pristine forward-power lift-off narrative.
E5.6 - severity race	Ask helix-vs-traction event ordering	568/572 completed: 0 helix-first, 30 simultaneous/unresolved, 18 traction-first, 520 neither.	Severe reversal is coupled to traction/hybrid transitions. Dynamics remain sizable before slip.
E5.7 - coupled chronology	Resolve clamp, traction, helix and dynamic terms through event	30 helix-crossing cases; 14 dynamic-only crossings; no $N_s=0$ at helix crossing.	Helix reversal can be dynamically driven, reduces secondary clamp, but is not belt contact loss.
E5.8 - paired performance	Run same transient with full vs quasi-static helix	Severe reversal pair materially diverges; stock control barely diverges.	This closes the chain from local helix mechanics to system-level trajectory consequence.



E1: quasi-static selected-flank zero-reaction boundary versus shift. This establishes the torque scale needed before transient dynamics are considered.

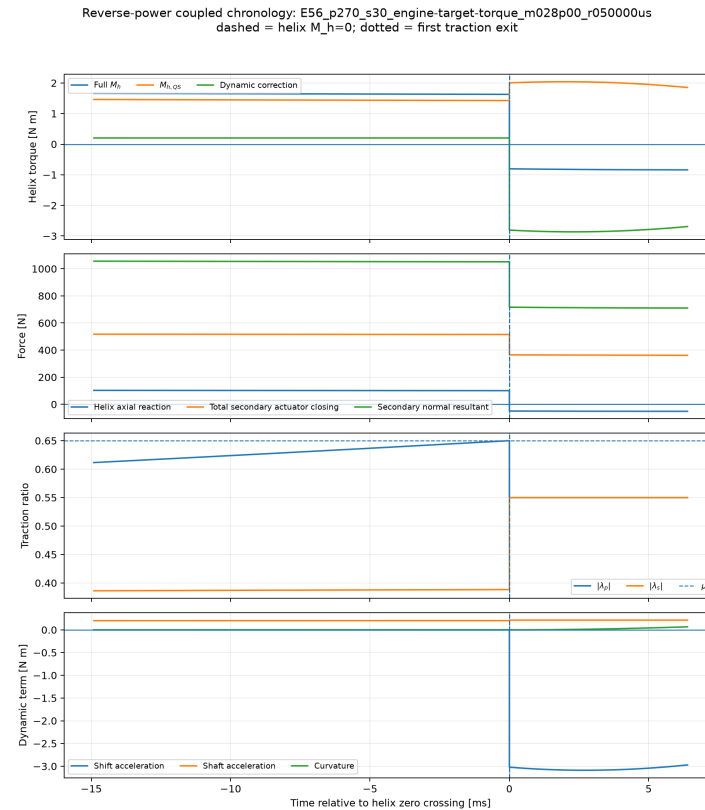
4. Severity race: dynamic terms exist before the belt leaves stick



E5.6. Left: event ordering by preload. Right: pre-traction shift-inertia contribution versus traction utilization. The study found no clean helix-first forward-power case, but it disproved the idea that appreciable dynamic helix terms only appear after slip.

Metric	E5.6 result
Planned / completed	572 / 568
Event ordering	0 helix-first; 30 simultaneous/unresolved; 18 traction-first; 520 neither
Most negative pre-traction total dynamic correction	-1.121 N m
Most negative pre-traction shift-acceleration term	-0.926 N m
Maximum pre-traction shift acceleration	16.66 m/s ²
Interpretation	Dynamic helix terms are real in stick-stick operation; the current hardware/preload simply leaves large positive margin under forward power.

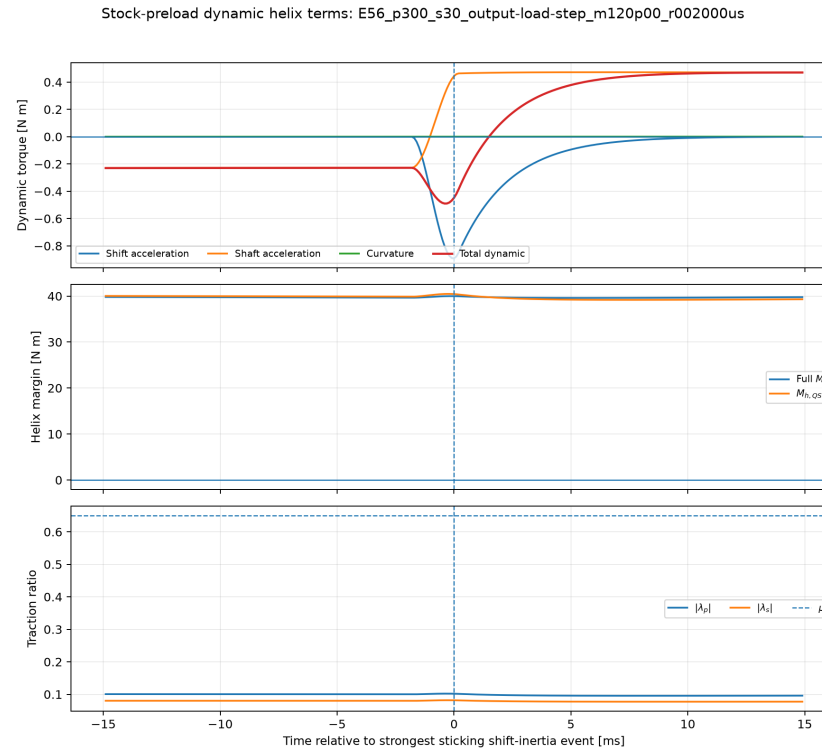
5. Coupled reversal chronology: what actually happens at the severe event



E5.7 selected reverse-power chronology. Dashed line marks helix zero crossing; dotted line marks first traction exit. The hybrid transition changes shift acceleration sharply, making the shift-inertia term dominate the dynamic correction.

Quantity	Selected E5.7 reversal exemplar
Case	E56_p270_s30_engine-target-torque_m028p00_r050000us
Preload / restart	270 deg / ~30% shift
QS margin at helix crossing	+1.811 N m
Dynamic correction at helix crossing	-1.811 N m
Secondary internal belt power at crossing	-5.00 kW
Helix-vs-traction event gap	-3.16 microseconds (treated as simultaneous/unresolved)
Minimum secondary normal at any helix crossing	827 N
Normal contact zero seen at crossing?	No
Dynamic-only crossings in E5.7 catalogue	14

6. Stock-preload control: dynamics can be appreciable without topology change

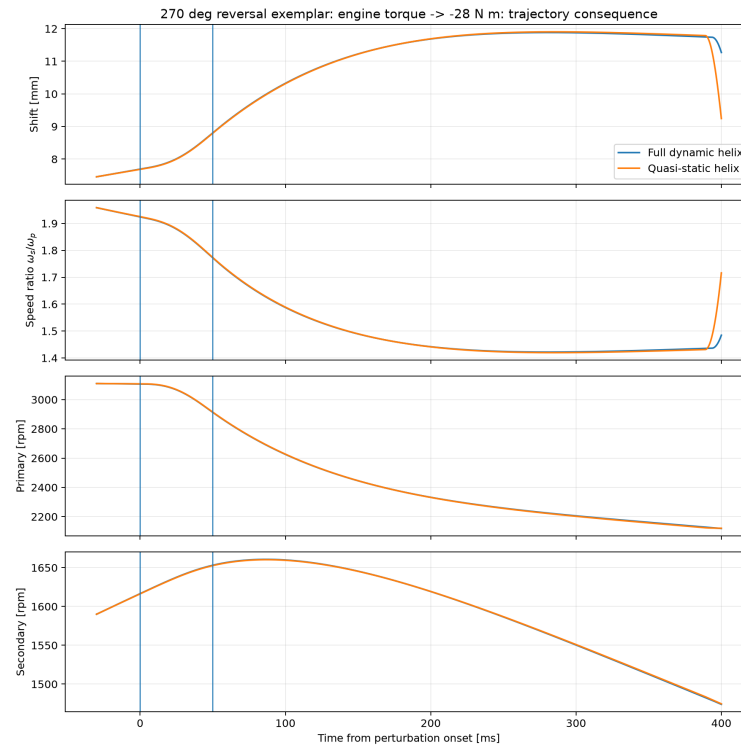


E5.7 stock 300 deg exemplar. The shift-acceleration term is visible while both contacts remain comfortably sticking, but the quasi-static margin is so large that no flank reversal occurs.

Metric	Stock E5.7 value
Case	E56_p300_s30_output-load-step_m120p00_r002000us
Shift-acceleration helix term	-0.891 N m
Shaft-acceleration helix term	+0.444 N m
Curvature term	1.46e-06 N m
Net dynamic correction	-0.448 N m
Full / QS margin	39.98 / 40.43 N m
Traction lambdas	primary 0.102; secondary -0.082
Secondary normal resultant	5857 N

Key interpretation: the dynamic terms are not artifacts that only appear after loss of traction. They can be order-1 N m corrections while the belt is sticking. Whether they matter to topology or system response depends on the available quasi-static margin and the transient severity.

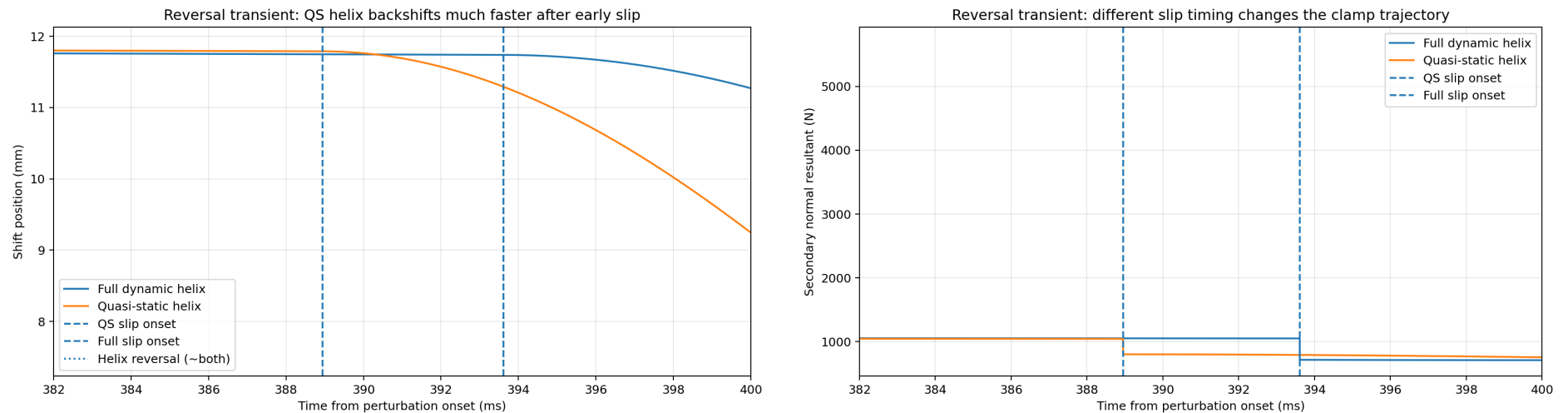
7. Paired full-vs-quasi-static helix: severe reversal performance



E5.8 severe reversal pair. The trajectories are almost coincident until the late hybrid event, then the quasi-static-helix solution backshifts much harder.

Performance metric	Full dynamic helix	QS helix	Difference / note
First non-stick time	393.610 ms	388.949 ms	QS exits stick 4.661 ms earlier
Non-stick duration	6.390 ms	11.051 ms	QS +4.661 ms
First helix reversal	393.606 ms	393.631 ms	Only 24.4 microseconds apart
Max shift difference			2.026 mm
Max speed-ratio difference			0.231
Max primary RPM difference			4.81 rpm
Max secondary RPM difference			1.04 rpm
Max secondary normal-force difference			258 N
Max secondary internal-power difference			620 W
Minimum secondary normal	710 N	756 N	Both remain in positive normal contact

8. Why the paired trajectories diverge: reflected inertia damps the backshift



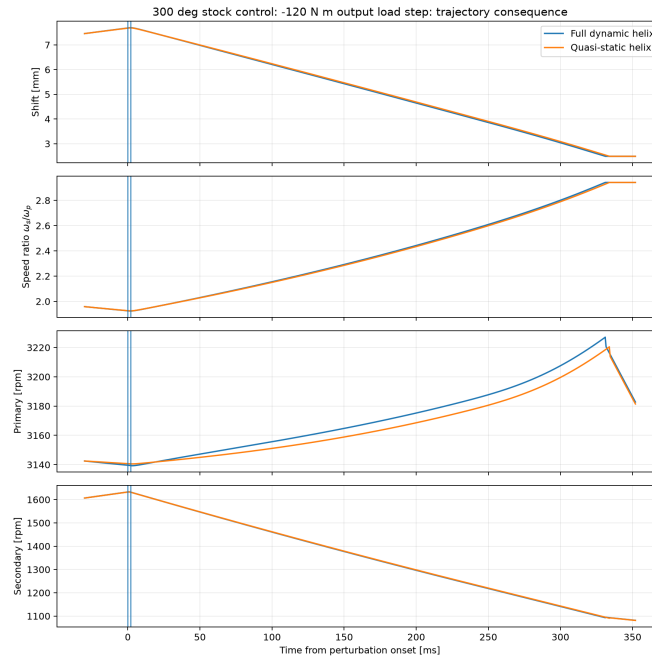
Event zoom. The quasi-static helix exits stick first and then backshifts more aggressively. The resulting clamp/normal-force histories separate as the two models follow different states.

Reflected-inertia anchor near the event

Quantity	Full dynamic	QS helix	Meaning
Helix-reflected active generalized shift mass	7.48 kg	0.00 kg	QS returns movable-member inertia to rigid shaft rotation; it does not delete mass.
Total direct active generalized shift mass	9.67 kg	2.17 kg	The full model has much greater resistance to rapid shift-coordinate acceleration.

This is the second major finding: the same movable-member inertia that contributes acceleration-dependent helix reaction also reflects through the helix kinematics into the shift coordinate. It therefore dampens the very rapid backshift that can generate a large dynamic reaction correction. The dynamic helix is two-sided, not simply an extra destabilizing load term.

9. Paired stock control: far from the boundary, trajectory consequences are small



E5.8 stock 300 deg control. Despite nonzero dynamic helix corrections, both paired trajectories remain very close in the ordinary/high-margin regime.

Metric	Paired stock result
Helix reversals	None in either model
Maximum shift difference	0.0514 mm
Maximum speed-ratio difference	0.0137
Maximum primary RPM difference	8.59 rpm
Maximum secondary RPM difference	2.45 rpm
Minimum full / QS helix margin	39.60 / 39.64 N m
Minimum secondary normal, full / QS	5816 / 5822 N
Important plotting note	Very large raw peak differences in force/power are narrow hybrid-event timing spikes. Do not present those maxima as sustained physical separation.

Interpretation: a quasi-static helix can be a good practical approximation when the dynamic correction is small relative to the available helix margin and the transient does not drive the system into a sensitive hybrid regime. The novelty is therefore regime-dependent, not universal.

10. What we can claim, what we infer, and what we should not overclaim

Category	Checkpoint statement
Demonstrated directly	Acceleration-dependent helix terms can be order-1 N m while the belt remains stick-stick.
Demonstrated directly	The full dynamic helix margin can become negative while the trajectory-frozen quasi-static margin remains positive; E5.7 catalogued 14 such crossings.
Demonstrated directly	The movable-member inertia reflects into the shift coordinate and materially changes the trajectory near a severe reversal event.
Demonstrated directly	Paired full-vs-QS runs produce different slip timing, shift/ratio histories, clamp histories and internal power histories in the severe reversal case.
Demonstrated directly	At stock/high-margin conditions the paired trajectory differences are small.
Physical interpretation	Reverse power collapses the quasi-static margin; a traction/hybrid transition can then create large shift acceleration, which magnifies the direct helix dynamic term. Reflected helix inertia simultaneously moderates that acceleration.
Physical interpretation	Helix unloading/reversal materially changes secondary closing force, but the studied helix crossings do not coincide with $N_s = 0$ belt contact loss.
Do not overclaim	Do not make helix lift-off itself the headline novelty. The stronger result is dynamic helix coupling and its system consequence.
Do not overclaim	Do not claim helix reversal clearly precedes traction loss in these severe cases; microsecond-scale interpolation differences are treated as simultaneous/unresolved.
Do not overclaim	Do not claim literature priority or uniqueness until the final literature comparison is performed.
Do not overclaim	E5.8 is full-dynamic slotted vs quasi-static slotted, not a physically detached unilateral comparator.

Preferred results framing

The secondary helix should be viewed as a dynamically coupled movable-member mechanism. Its inertia enters both the instantaneous reaction law and the generalized shift inertia. These effects can be negligible in high-margin operation yet become trajectory-defining near severe reverse-power / traction transitions. Helix flank reversal is one visible consequence of that broader dynamic coupling.

11. Restart guide: where we left off

Item	Checkpoint
Current status	Helix discovery is complete enough for synthesis/write-up. Do not launch another broad sweep by default.
Headline severe case	E58_reversal_270_s30_engine_m28_r050 (paired E5.8); parent E5.7/E5.6 case E56_p270_s30_engine-target-torque_m028p00_r050000us.
Stock control	E58_stock_300_s30_output_m120_r002; parent E5.7 case E56_p300_s30_output-load-step_m120p00_r002000us.
Primary evidence chain	E5.6 establishes pre-slip dynamic terms -> E5.7 resolves coupled reversal chronology -> E5.8 proves trajectory-level consequence with paired models.
Best headline figures	E5.7 reversal_coupled_chronology; E5.8 severe trajectory_performance; event zoom shift/normal force; E5.8 stock trajectory_performance.
Optional future run	Only a small 2-3 case neighborhood around the severe E5.8 exemplar if final manuscript figures need robustness evidence.
Later model question	A true unilateral detached helix requires releasing both contact force and $\theta = \theta(s)$ kinematic constraint and adding an independent movable-member relative-angle DOF. Do not use force clipping with the constraint retained as the primary physical comparator.

Artifact provenance used for this checkpoint

- helix_topology_discovery_artifacts(3).zip - E1 through E5.5 summaries and figures.
- helix_topology_discovery_artifacts(5).zip - E5.6, E5.7 and E5.8 summaries, paired traces and performance figures.
- Study tree: results/cinder-v1.1.2/studies/helix-topology/.
- Comparator: tagged CINDER quasi_static_helix ablation; dynamic coupling removed while movable-sheave inertia is returned to rigid secondary shaft inertia.

Bottom line for future us

The study succeeded because it established more than a lift-off event. It showed that secondary helix dynamics act through both reaction-force corrections and reflected shift inertia, that these mechanisms can oppose/moderate each other, and that their net effect can propagate into measurable differences in ratio, RPM, slip, clamp and power trajectories. Lift-off is a useful boundary case; dynamic helix coupling is the result family.